5(a)	Any 2 from: • noise can be filtered out / noise can be removed / signal can be regenerated • can carry more information per unit time / greater rate of transmission of data • can have extra bits of data to check for errors • can be encrypted	**B2**
5(b)(i)	$v \propto \lambda$	**C1**
	ratio $= v_{air} / v_{fibre}$ $= 3.00 \times 10^8 / 2.07 \times 10^8$ = **1.45**	**A1**
5(b)(ii)	attenuation = 10 log (P_2/P_1)	**C1**
	$0.40 \times L = 10 \log (1.5 / 0.06)$ $0.40 \times L = 13.979$	**C1**
	L = **35** km	**A1**